

Project Completion Report

Project Name	Cardano Token Engineering Lab — Understanding and Leveraging Token Design to Grow Cardano
Project Number	1100073
Project Manager	Miguel Saldana
Project Start Date	January 2024 (Fund11 project onboarding)
Project Completion Date	8/7/2026

Project Summary

The Cardano Token Engineering Lab (CTEL) was funded under Project Catalyst Fund11 to bootstrap a community resource for token engineering on Cardano: educational research, case studies of live Cardano DeFi protocols, and open-source tooling to pull on-chain data for systems modeling. The project was delivered across three milestones, summarized below.

#	Milestone	Budget	Delivery	Status
1	Research, data, and curation of data in a stored repository	6,000 ADA	Month 1 — Apr 2024	Completed
2	Case Studies and Partnership on/with Cardano DeFi Protocols	6,000 ADA	Month 3 — Jun 2024	Completed
3 (Final)	Code Framework for Modeling Data Acquisition	8,000 ADA	Month 4 — Jul 2024	Completed (this submission)

1. Deliverables

Milestone 1 — Research, Data, and Curation of Data in a Stored Repository

A publicly available repository of project research and material was established as the foundation for the Lab's later case studies and code framework.

- Repository: <https://github.com/Cardano-Token-Engineering-Lab/Home>

Milestone 2 — Case Studies and Partnership on/with Cardano DeFi Protocols

Three in-depth research papers were produced, each covering token design, mechanism design, incentive models, and economic security analysis for a protocol in a distinct DeFi category (lending/borrowing, CDP/synthetic assets, and AMM/DEX), satisfying the milestone's requirement to cover at least one protocol from each category among the top 20 most active Cardano DeFi projects.

- Liquid Finance (lending/borrowing) — token design, incentive mechanisms, and economic security analysis

- Indigo Protocol (CDP / synthetic assets) — peg-stability mechanism evolution and economic security analysis
- Minswap (AMM / DEX) — LP fee governance and economic security analysis

Papers are hosted in the Lab's research repository: <https://github.com/Cardano-Token-Engineering-Lab/Research> — [\[LINK\]](#). Each paper includes a numbered, footnoted bibliography and is available in both Word and PDF format.

Formal protocol and cross-ecosystem partnerships (the milestone's other stated goal) were not finalized as of this submission. Placeholder partnership-agreement files are maintained in the papers repository, to be completed if and when a formal partnership is reached.

Milestone 3 (Final) — Code Framework for Modeling Data Acquisition

A parameterized, tested Python framework was delivered for pulling Cardano native-token on-chain data into pandas DataFrames for systems modeling, replacing the project's earlier token-specific prototype scripts.

- Repository: <https://github.com/Cardano-Token-Engineering-Lab/Query-Cardano-Data>
- Framework: `cardano_token_framework`, a common `TokenDataSource` interface implemented by both Blockfrost and Koios backends, covering asset info, holders, transactions, and swap-type activity
- CLI and library access: `python -m cardano_token_framework`, plus direct Python imports
- Test suite: 51 tests, 100% statement coverage, run via mocked API clients (no live network dependency)
- Linting: `ruff`, zero errors
- CI: GitHub Actions running lint + tests on every push/PR across Python 3.10–3.12
- Example token model: a basic cadCAD model (`examples/cadcad_basic_model.py`) consuming the framework's own swap-volume output as an input signal
- Indexing research: `docs/indexing_roadmap.md` — a sourced, phased plan for adopting Dolos, Carp, or Oura/Scrolls as a more robust indexing backend

Open source status: Yes. Licensed under MIT (see LICENSE in the repository). GitHub is the primary and permanent host for all project outputs across all three milestones.

Testing performed: automated unit testing (pytest) with 100% statement coverage on the framework package, plus a live end-to-end run of the example cadCAD model, both re-run and verified at the time of this submission — see `POA_EVIDENCE.md` in the repository for captured command output tied to each acceptance criterion.

2. Usage

As a public good aimed at Cardano developers, researchers, and token designers, the primary users of this project's outputs are:

- Cardano DeFi protocol teams referencing the Liqwid, Indigo, and Minswap research papers for economic security review
- Developers and researchers using `cardano_token_framework` to pull token data for their own systems models
- The wider Cardano Token Engineering community, via the Lab's GitHub organization as an ongoing public resource

Engagement evidence: Usage tracking is planned as this is a newly built repository

3. Impact

This project delivered the first open-source, Cardano-specific token engineering research and tooling public good of its kind: three fully sourced protocol research papers covering the three major categories of Cardano DeFi mechanism design, and a reusable, tested code framework that lowers the barrier for anyone to pull Cardano token data into a Python-based systems model.

- Before: no consolidated, open-source repository of Cardano DeFi economic security analysis existed comparable to what other ecosystems (e.g., Token Engineering Academy, BlockScience) maintain.
- After: three protocols now have public, citable economic security research, and the underlying data-acquisition problem, the main practical barrier to producing more such research, has a tested, reusable solution.

Quality proof: All teams were reached out to for comments about the research papers, but only Florian Valery from Liqwid hopped on a call and provided commentary and feedback.

Cardano ecosystem benefit: strengthens the Cardano DeFi ecosystem's collective understanding of token design risk and provides a reusable data-acquisition layer other researchers and protocol teams can build on directly rather than re-implementing from scratch.

4. Sustainability

This is intended as an ongoing effort rather than a one-off deliverable.

- Maintenance model: the Cardano Token Engineering Lab GitHub organization remains the permanent, publicly accessible home for all project outputs. Both repositories (research papers and the code framework) are open source under MIT license and accept community contributions and issues.
- Revenue/continuation model: The sustainability model will be made up through paid token engineering consultations, devoting a portion of the fees to maintaining and update the library and information. In lieu of active projects, it will be maintained by myself through volunteer efforts.
- Future roadmap: `docs/indexing_roadmap.md` in the Query-Cardano-Data repository lays out a phased plan for extending the framework (self-hosted Dolos backend, true swap-decoding via Carp, and p2p/staking transaction support) beyond what this milestone delivered.

Forking / permanent storage: both repositories are permanently hosted on GitHub under the Cardano Token Engineering Lab organization and are fully open source (MIT), so any community member may fork, extend, or archive them independently of the Lab's continued involvement.

Project Completion Video

Video link: [Project Closeout Video on YouTube](https://youtu.be/gYteWhakEwk) (<https://youtu.be/gYteWhakEwk>)

The video covers: the challenge entered and why, the approach proposed and funded, a live technical walkthrough of the delivered code framework and research papers, progress and challenges encountered across the three milestones, and next steps for the project.

Before submitting, verify every link in this report resolves correctly — a broken link can cause the PCR to be rejected.